Fiberwise Truncations, Equifibered Squares, and the Freudenthal Suspension Theorem

John Wheeler

November 12, 2025

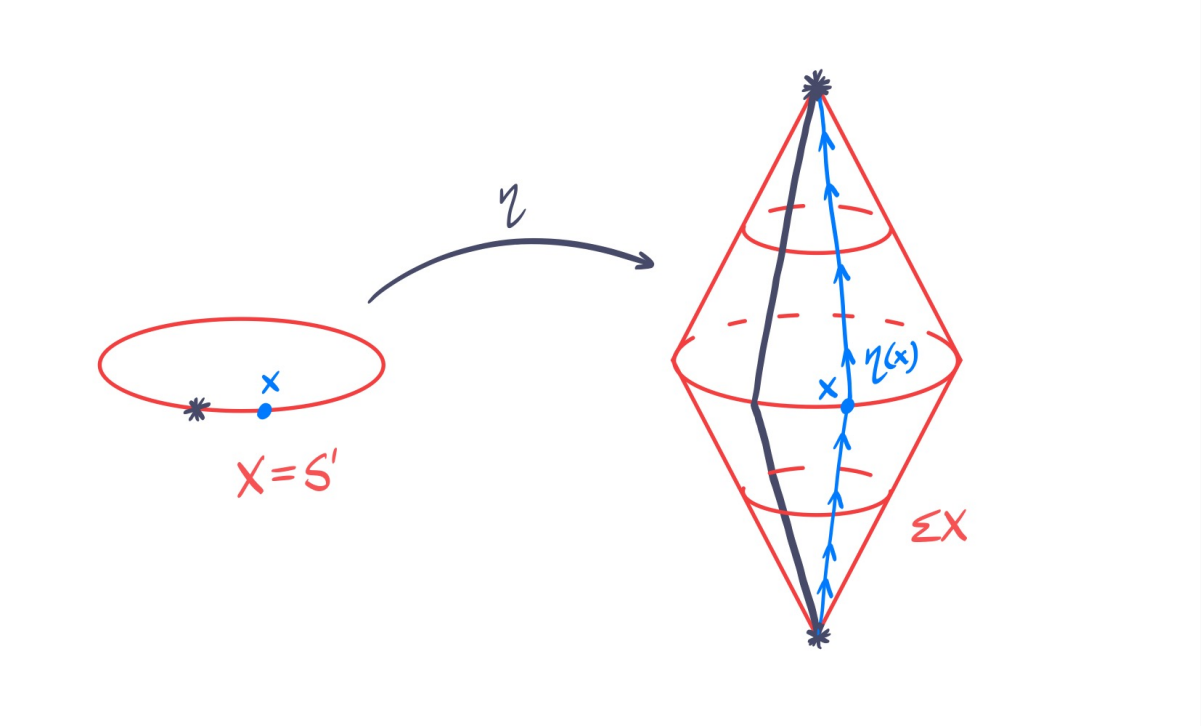
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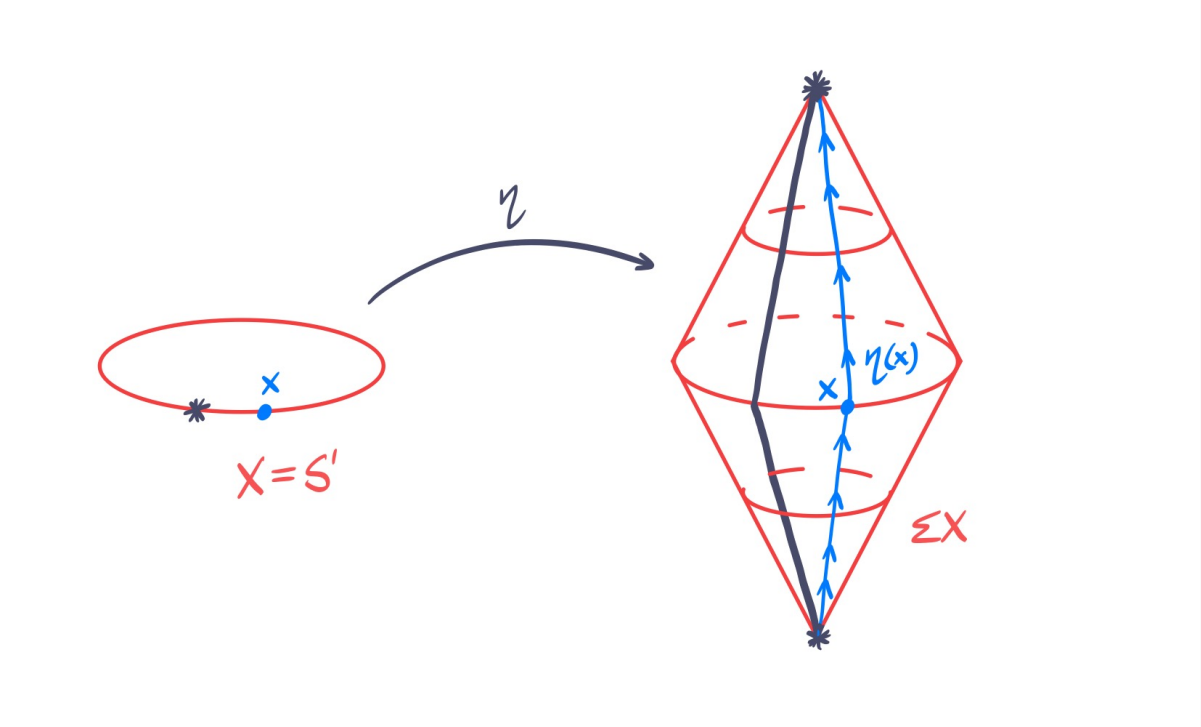
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Warning: We use the indexing conventions of Rezk. What we call an n-connected map is what is called an $(n-1)$-connected in most references.

n-Connected & n-Truncated Maps: Definition

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Definition. A map $f : X \to Y$ is n-**connected** if $\forall x \in X$

$$f_* : \pi_k(X, x) \to \pi_k(Y, f(x))$$

is an isomorphism for $k \leq n$ and an epimorphism for $k \leq n+1$. A space X is n-**connected** if the terminal map $* : X \to \mathrm{pt}$ is n-connected.

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Definition. A map $j : A \to B$ is n-**truncated** if $\forall a \in A$

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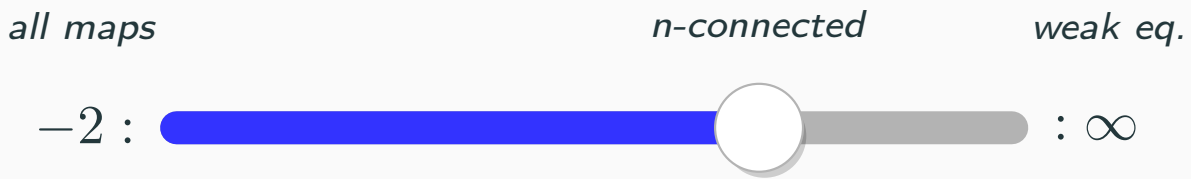
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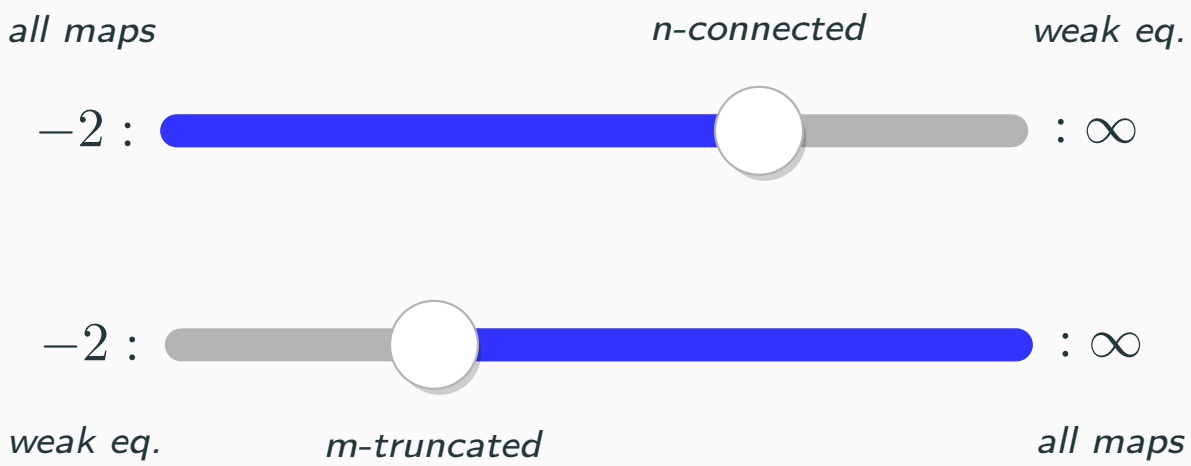
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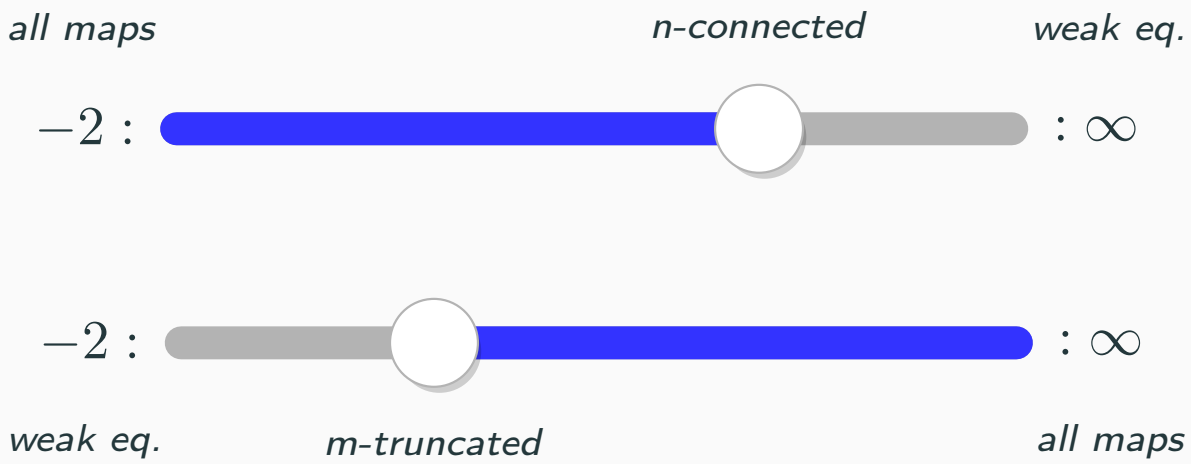
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4. A map that is n-connected *and* m-truncated with $n \geq m$ is a weak equivalence



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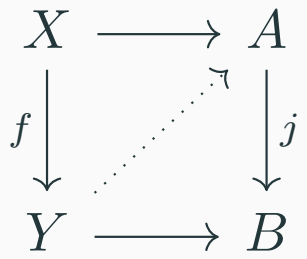
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$$\begin{array}{ccc} X & \longrightarrow & A \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle j} \\ Y & \longrightarrow & B \end{array} \qquad (\bigstar)$$

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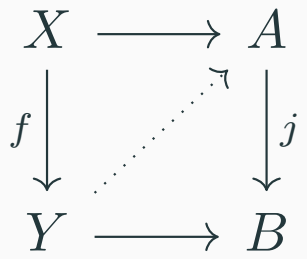


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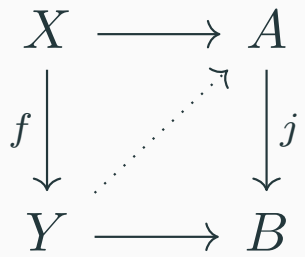
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Lemma. A map $f : X \to Y$ is n-connected iff $\forall\ j : A \to B$ n-truncated, $\forall$ (★) has a unique solution up to homotopy.

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Remark. These results, and the Freudenthal suspension theorem, are usually proved as corollaries of Blakers-Massey, but we can prove them formally and directly without Blakers-Massey.

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Proposition/Definition. For all $n \geq -2$, $\forall$ maps $f : X \to Y$ there exists a factorization

$$X \xrightarrow{i} |f|_n \xrightarrow{j} Y$$

with i n-connected and j n-truncated. This factorization is unique up to homotopy, and is called the **fiberwise n-truncation** of the map f.

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Definition. A commutative square is n-**equifibered** if after taking fiberwise n-truncations of f and f'

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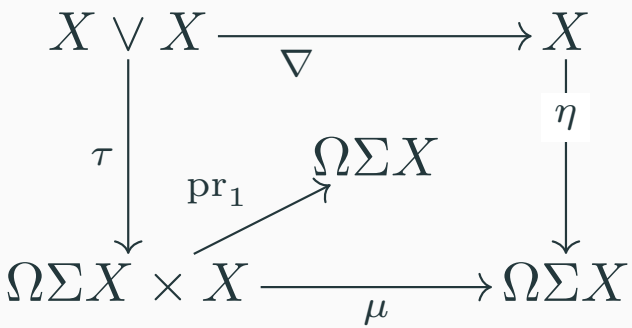
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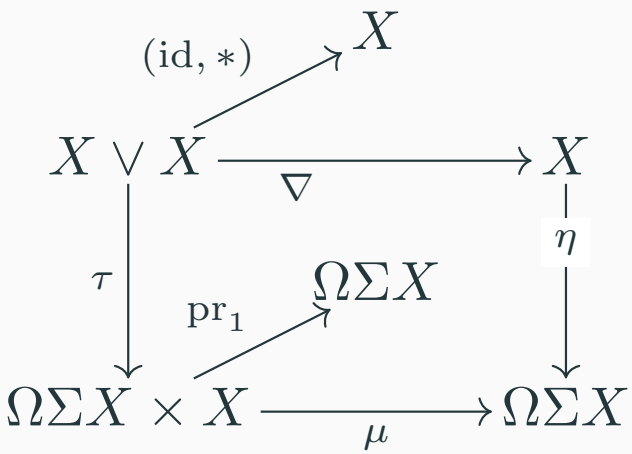
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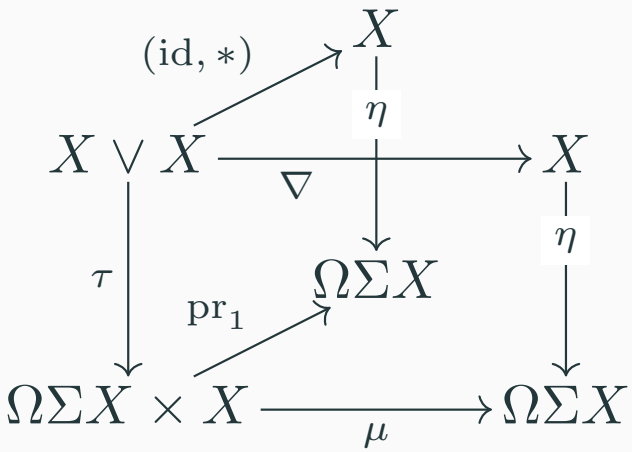
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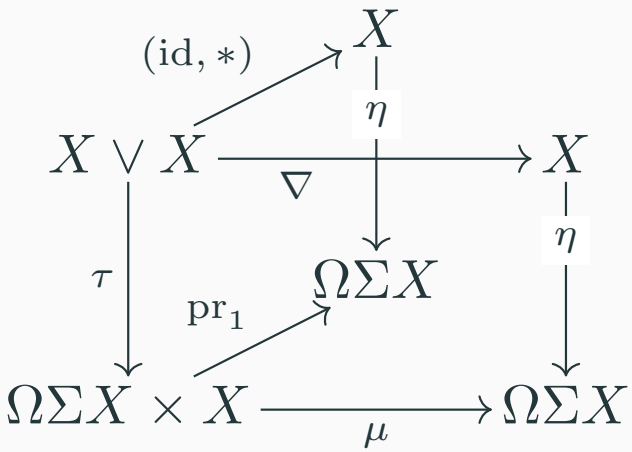
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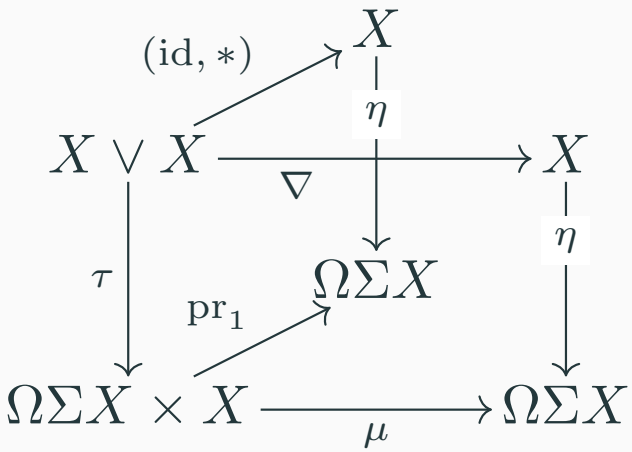
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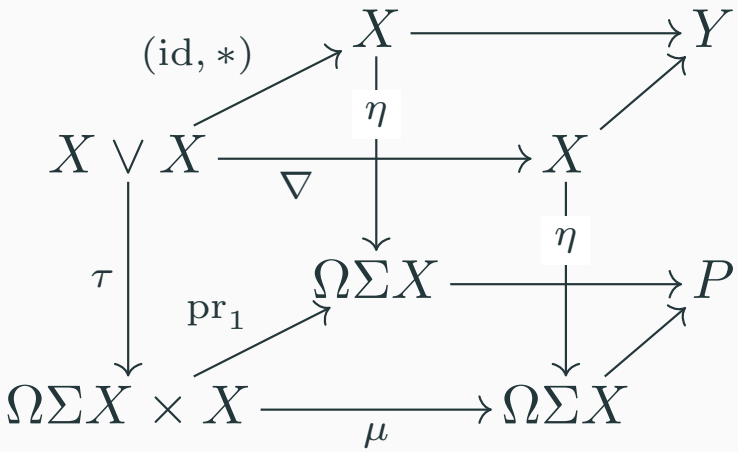
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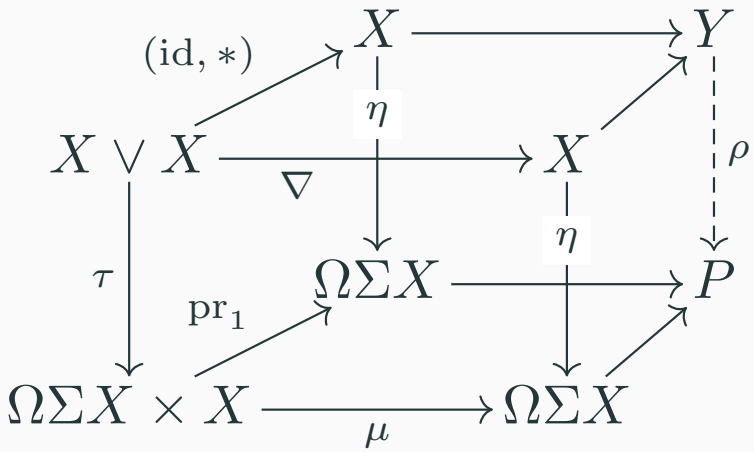
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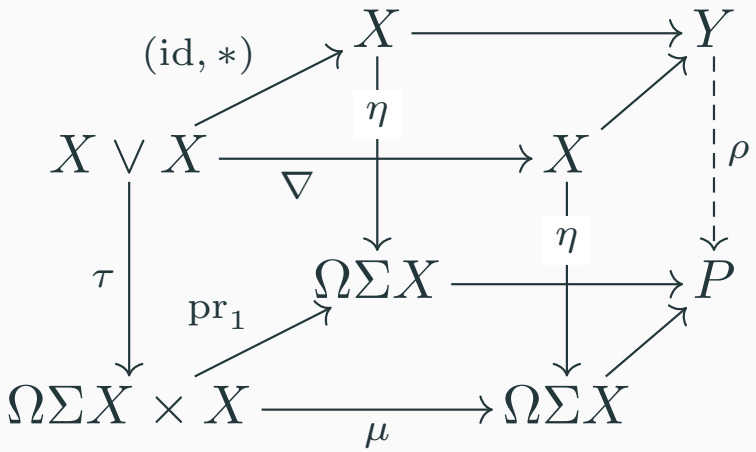
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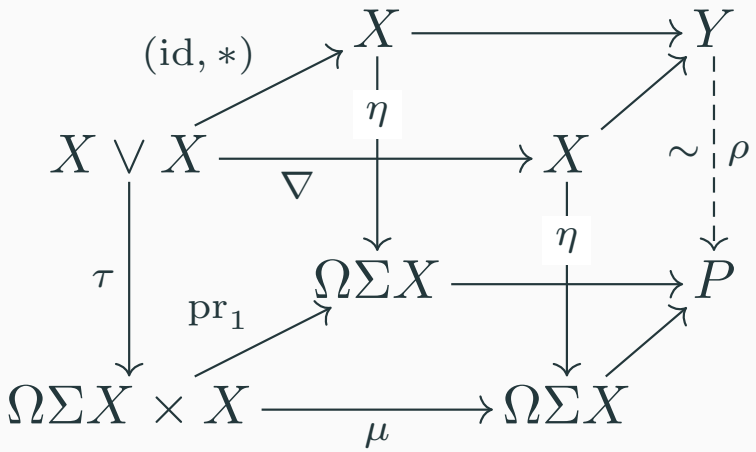


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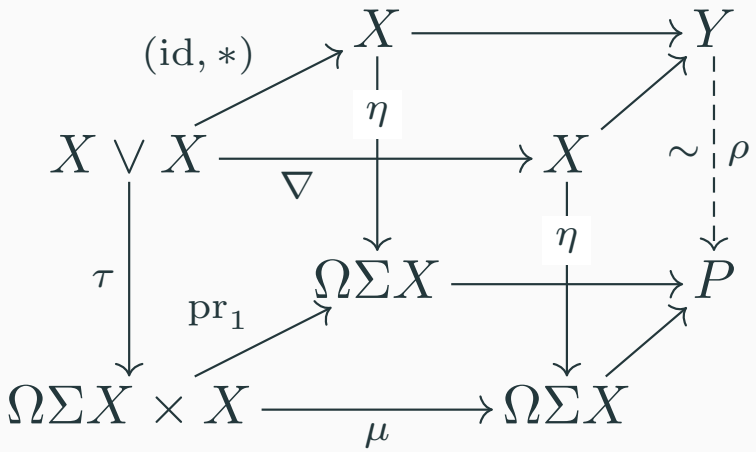


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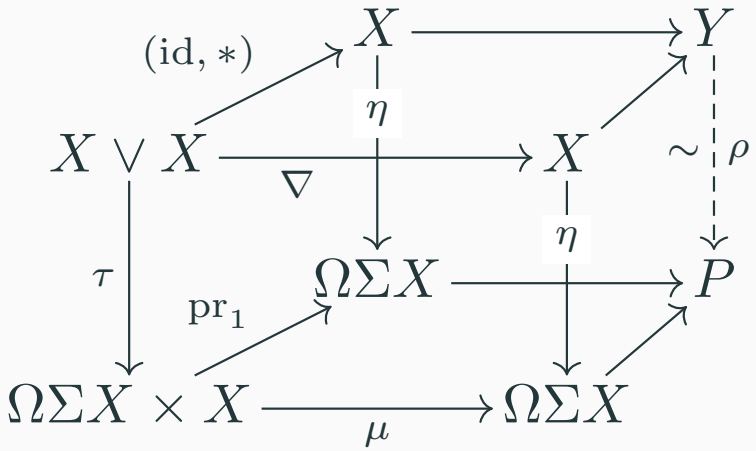
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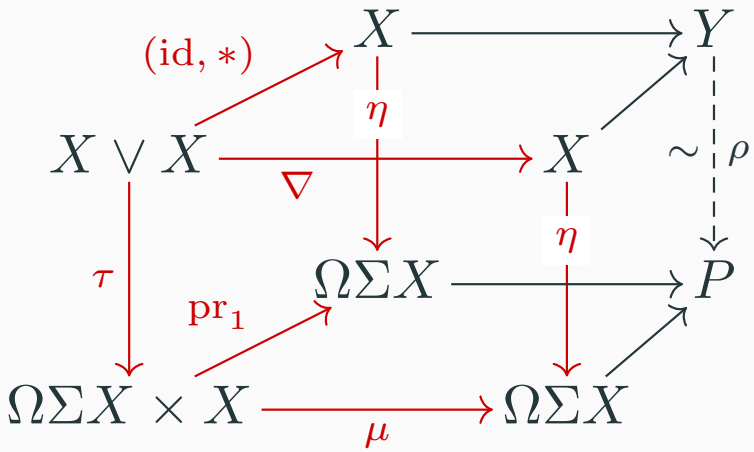


By inspection Y and P are contractible $\implies$ ρ is an equivalence

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The Freudenthal Suspension Theorem: Proof Sketch

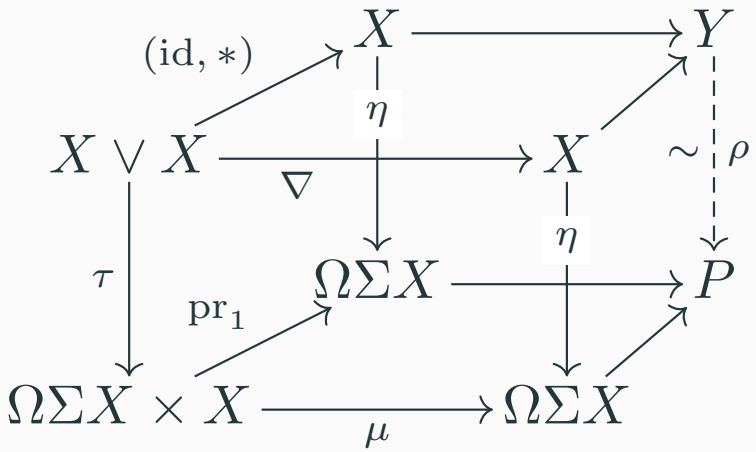
Let X be pointed and n-connected. **Claim.** $\eta : X \to \Omega\Sigma X$ is $(2n)$-connected.



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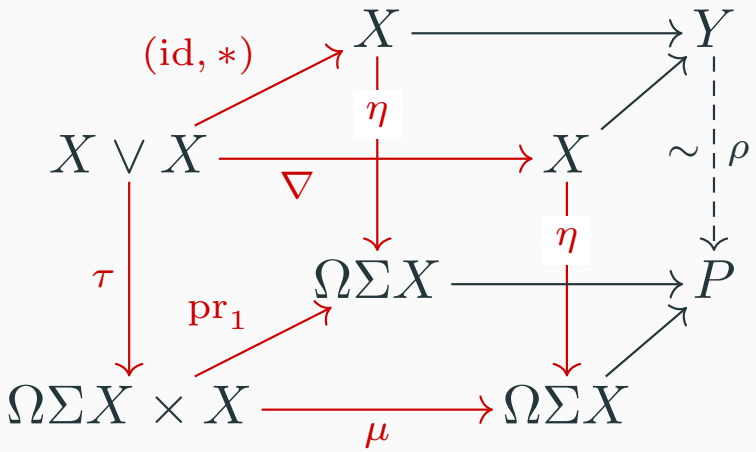


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Lemma (Patching). Consider a commutative cube oriented as above. If the front and left squares are m-equifibered and the top and bottom squares are homotopy pushouts, then the back and right squares are m-equifibered.

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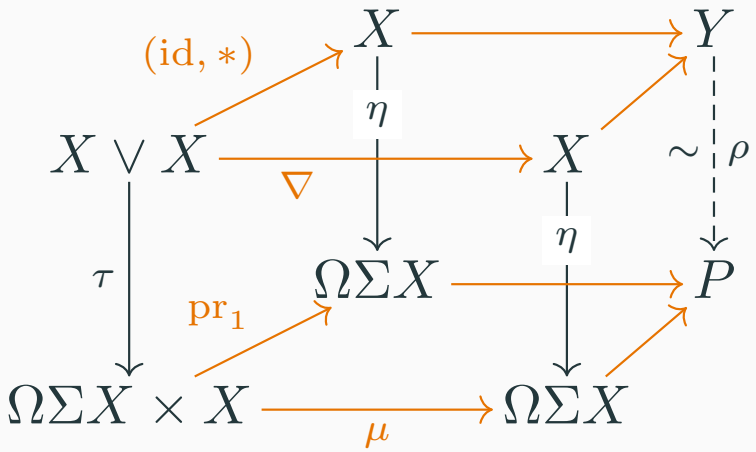


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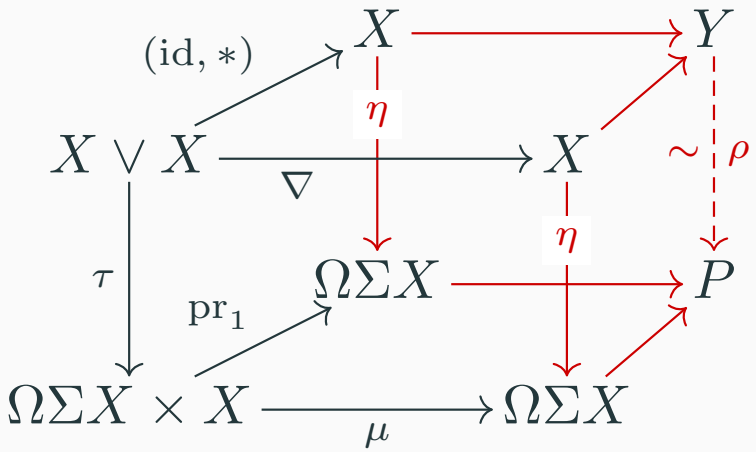


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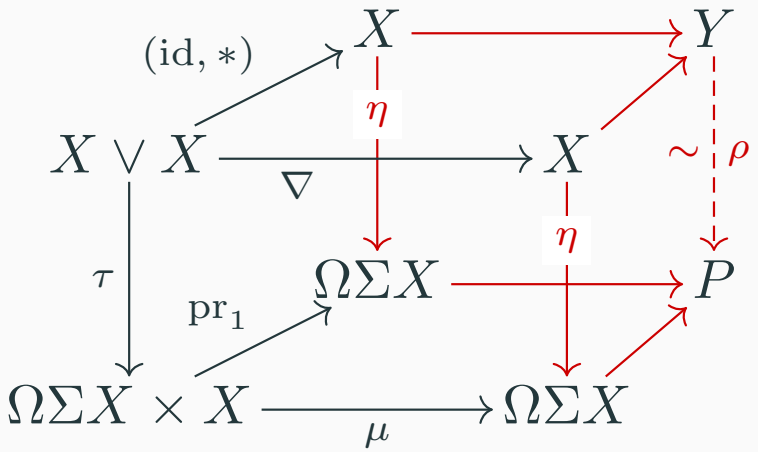


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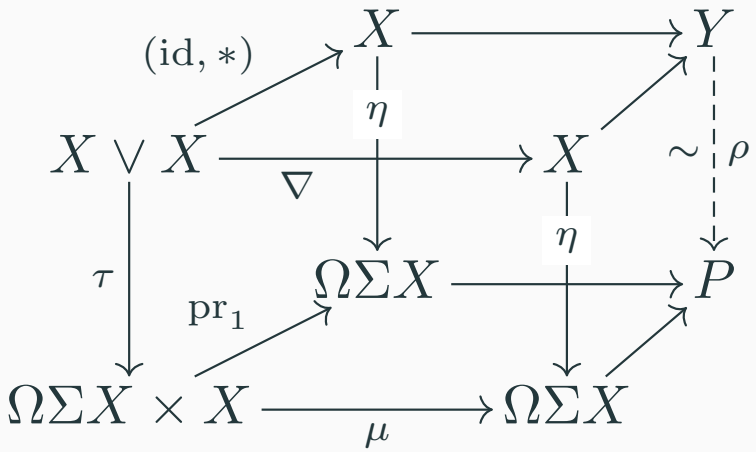


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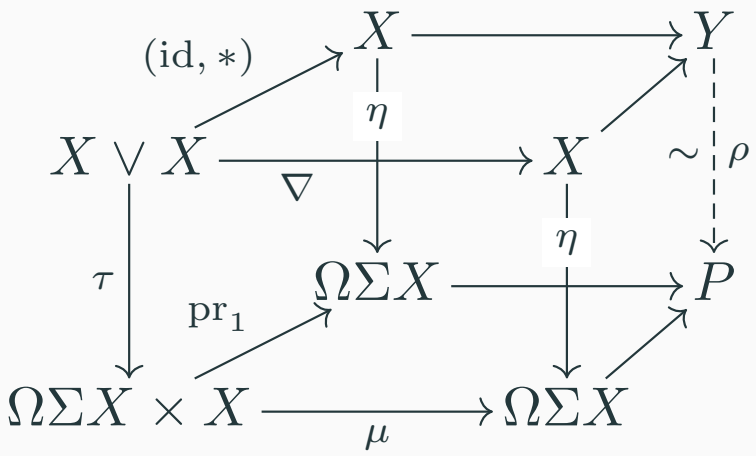
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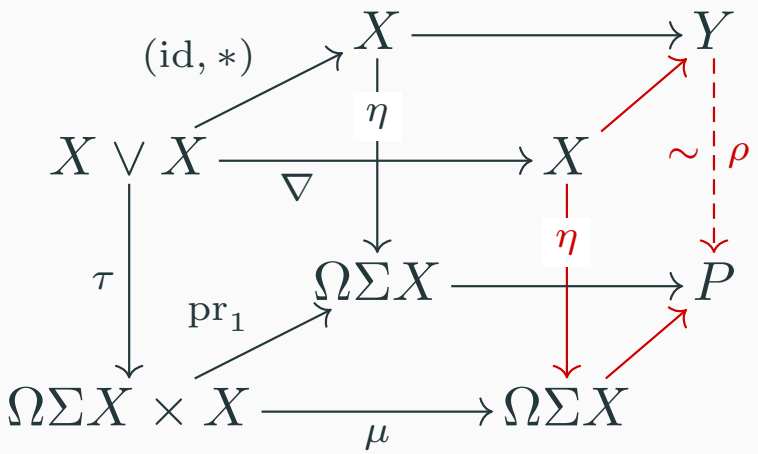


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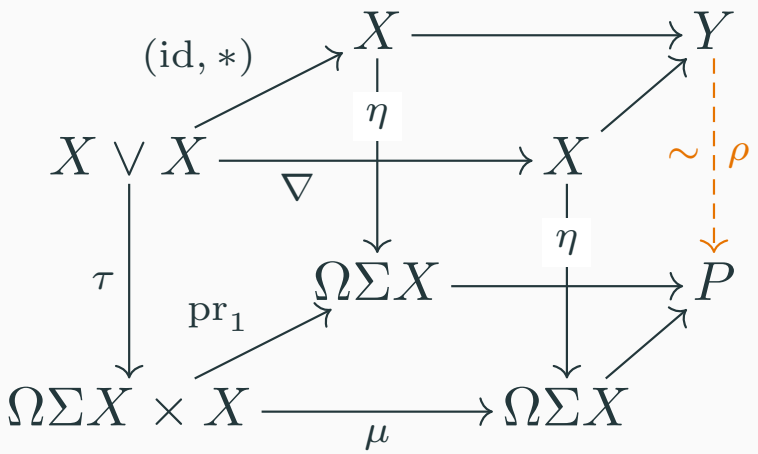


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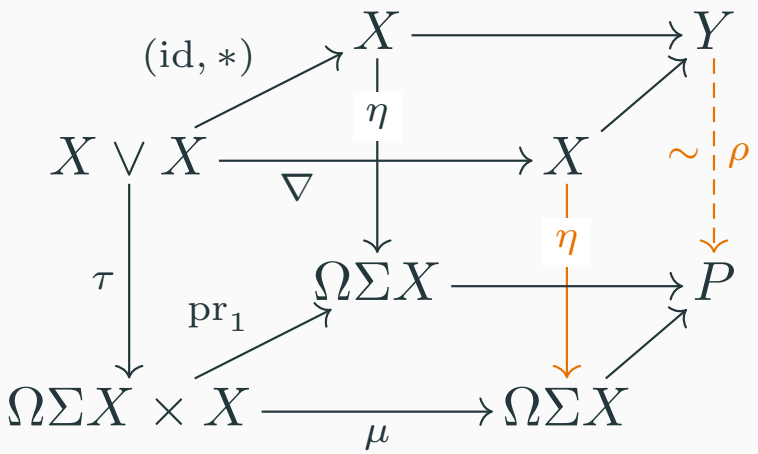


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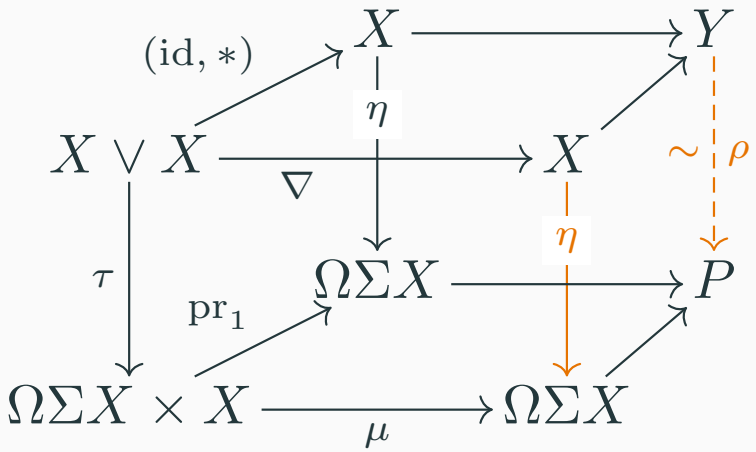


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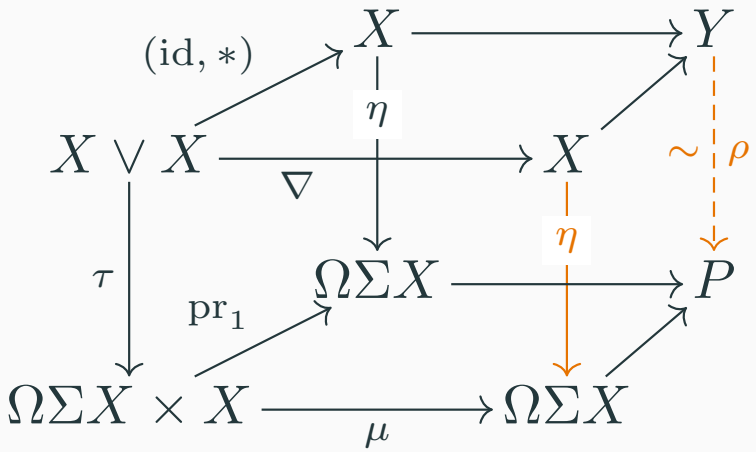


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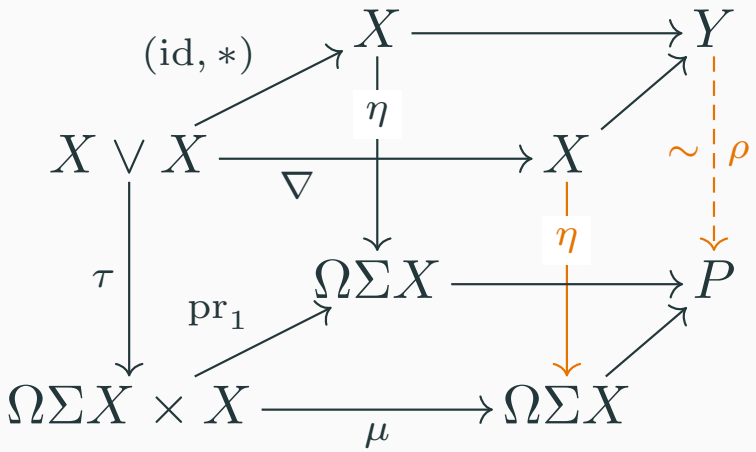
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References

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